Armaan Mathur

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Summary

Nanotechnology Engineering student at the University of Waterloo focused on **digital hardware design and verification**. Experienced in **SystemVerilog RTL development, FPGA design**, and testbench verification through ASIC club work.

Education

University of Waterloo **Sep. 2025 – Expected 2030**
Bachelor of Applied Science in Nanotechnology Engineering *Waterloo, ON*

Experience

RTL Design & Verification Member **April 2026 – Present**
ASIC Chip Design Team (UWASIC) *University of Waterloo*

- Developing synthesizable **SystemVerilog RTL** for a modular **10 Gb/s Ethernet** PCS, implementing and integrating a 64b/66b transmit encoder that maps 64-bit XGMII data and 8 control bits into standards-compliant 66-bit blocks.
- Implementing datapath and control logic for **14 control-block** formats, including data, idle, ordered-set, Start in 2 legal byte lanes, Terminate across all 8 positions, and deterministic handling of invalid sequences.
- Building a **self-checking encoder–decoder** verification environment with directed, edge-case, and malformed-packet tests; debugging simulation waveforms to validate sync headers, bit ordering and control code mapping.
- Collaborating on PCS subsystem integration with the **scrambler, gearbox, decoder, and MAC interface**, reviewing RTL and resolving protocol, byte-alignment, and module-interface issues within a Git-based development workflow.

Design Engineering Intern **May 2026 – Present**
Ontario Power Generation *Pickering, ON*

- Reviewed **electrical vendor schematics** and engineering packages for CNSC compliance, identifying critical design discrepancies and supporting acceptance sign-off for the **$26.8B Pickering Refurbishment project**.
- Performed load calculations and VFD harmonics studies, directly contributing to design change packages for Units 5–8 Retube, Feeder, and Boiler Replacement.
- Coordinated with external vendors and cross-functional teams during design reviews, resolving technical issues and accelerating sign-off on safety-critical records.

Projects

SPI-Controlled PWM Peripheral | *SystemVerilog, cocotb, Icarus Verilog, GTKWave* **March 2026**

- Designed a synthesizable **SystemVerilog SPI Mode 0 peripheral** with memory-mapped control of **16 digital outputs and 16 PWM channels**; verified register, reset, and PWM behavior using Python and cocotb.
- Built Python/cocotb testbenches covering SPI register writes, PWM duty-cycle boundaries, reset behavior, invalid transactions, and clock-domain crossings between **100 kHz SPI and 10 MHz system clocks**.
- Automated simulation checks with expected-value comparisons and debugged failures using GTKWave; validated the design through linting and synthesis.

Radiation–Tolerant FPGA Controller | *SystemVerilog, cocotb, Vivado, Basys 3* **August 2026**

- Designed an 8-bit FPGA telemetry controller with **Triple Modular Redundancy (TMR)** across 3 stateful processing cores, using bitwise majority voting to maintain correct outputs during modeled single-event upsets (SEU)
- Implemented **synthesizable fault injection**, replica-mismatch detection, and automatic state recovery, restoring a corrupted core from majority-voted state within 1 clock cycle.
- Built self-checking cocotb verification covering **100+ injected state faults** across all 3 replicas, achieving 100% masking and recovery for single-replica faults while validating expected failure under multi-replica corruption.

Python RTL Regression Tool | *Python, Icarus Verilog, Yosys, JSON* **July 2026**

- Developed a configuration-driven **Python RTL regression tool** that compiles SystemVerilog designs and executes **60+ automated simulations** across 3 RTL modules using Icarus Verilog.
- Automated **20+ directed and parameterized tests** with Markdown/JSON reporting and correctly detected **10/10 seeded RTL defects** involving reset, arithmetic, timing, overflow, and underflow behavior.

Technical Skills

- **Programming & Verification:** Python, cocotb, SystemVerilog Testbenches, MATLAB, JavaScript, Git
- **Digital Design & HDL:** SystemVerilog, Verilog, FPGA Design, Digital Logic, FSMs
- **Interfaces & Protocols:** Ethernet, SPI, PWM, Clock Domain Crossing; Familiar with (I2C, UART, AXI-Stream)
- **Simulation & EDA Tools:** Vivado, Icarus Verilog, GTKWave, Yosys
- **Hardware & Engineering:** Altium Designer, KiCad, PCB Layout, Circuit Analysis, Simulink, SolidWorks